Cardiovascular Diabetology

RESEARCH

Open Access

# Post-translational modifications in the insoluble fraction of platelet-poor plasma distinguish pre-COVID-19 Postural Orthostatic Tachycardia Syndrome, Long COVID, and Long COVID-POTS: a preliminary study

Renata M. Booyens[1*], Mare Vlok[1,2], Cecile Bester[3], Rashmin Hira[4], M. Asad Khan[5], Douglas B. Kell[1,6], Satish R. Raj[4*] and Etheresia Pretorius[1,6*]

## Abstract

**Background** Pre-COVID-19 Postural Orthostatic Tachycardia Syndrome (PC-POTS), Long COVID with initial orthostatic hypotension (LC-IOH), and Long COVID-POTS (LC-POTS) are primarily chronic post-viral conditions with debilitating symptoms, despite routine laboratory findings often being normal. We previously identified amyloidogenic fibrinaloid microclot complexes (FMCs) in the insoluble fraction of platelet-poor plasma (PPP-IF) in Long COVID, but it remains unclear whether similar abnormalities occur in PC-POTS and LC-POTS, or whether post-translational modifications (PTMs) contribute to disease mechanisms.

**Methods** Platelet-poor plasma from healthy controls ($n = 18$), PC-POTS ($n = 20$), LC-IOH ($n = 19$), and LC-POTS ($n = 18$) participants underwent proteomic analysis of the PPP-IF using double trypsin digestion and data-independent LC-MS/MS. Differential protein abundance, PTMs, and amyloidogenicity were compared across groups.

**Results** Although few proteins were significantly dysregulated, extensive disease-specific PTM alterations were identified across coagulation, immune, and metabolic pathways. LC-IOH demonstrated prominent AGE- and oxidation-related PTMs on fibrinogen subunits, particularly fibrinogen alpha chain (FIBA), resembling diabetic glycation profiles. PC-POTS showed fewer fibrinogen PTMs but marked metabolic and immune dysregulation, including oxidised apoA1 and apoB and complement-associated proteins (C3, C4A/B, IC1), IGG1, and A2MG. LC-POTS shared coagulation-related features with LC-IOH and immune-related features with PC-POTS. Many dysregulated

*Correspondence:
Renata M. Booyens
renatab@sun.ac.za
Satish R. Raj
satish.raj@ucalgary.ca
Etheresia Pretorius
resiap@sun.ac.za

Full list of author information is available at the end of the article

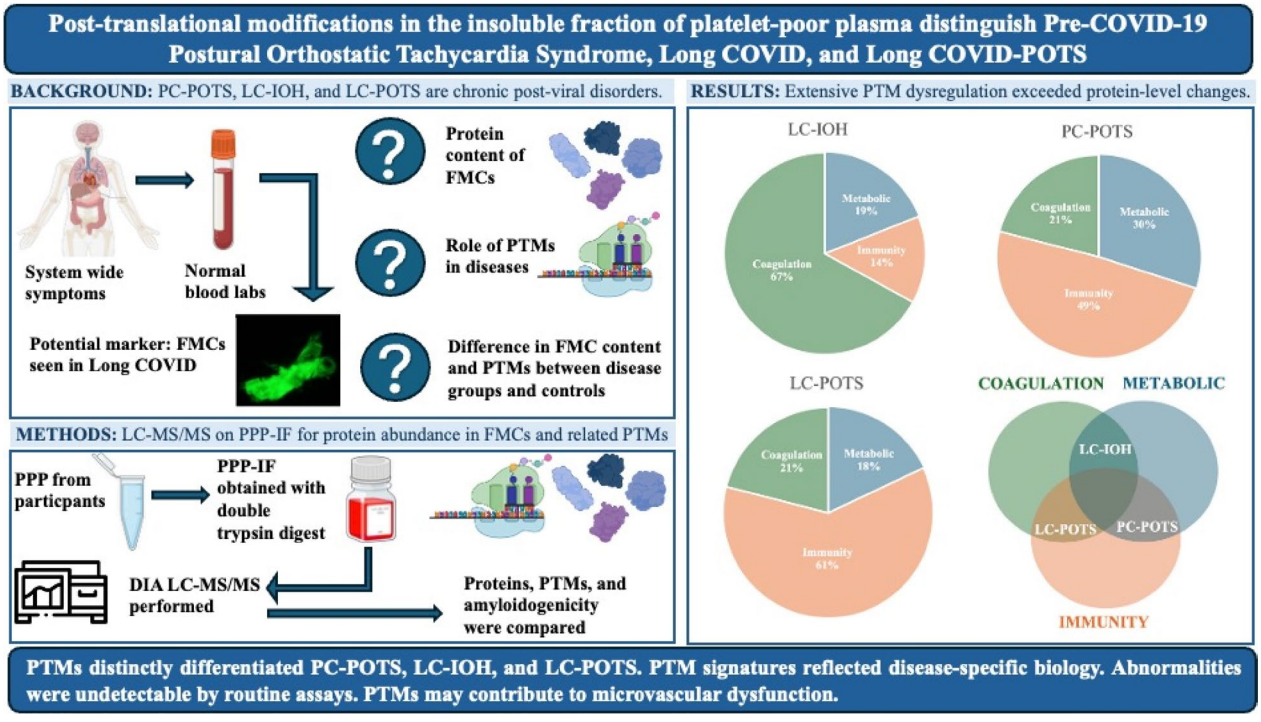
© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

peptides were highly amyloidogenic in silico, consistent with β-sheet-rich FMCs. PTM dysregulation was substantially greater than protein-level changes.

**Conclusions** This study provides the first evidence that PTMs within the PPP-IF has the potential to distinctly differentiate PC-POTS, LC-IOH, and LC-POTS. Because PC-POTS samples predated SARS-CoV-2, their PTM signatures reflect intrinsic disease biology independent of COVID-19. PTM profiling revealed pro-coagulant fibrinogen changes in LC-IOH and LC-POTS, metabolic-oxidative dysregulation in LC-IOH and PC-POTS, and immune dysregulation in PC-POTS and LC-POTS. These abnormalities were not detectable by routine assays and may contribute to microvascular dysfunction. The findings support disease-specific biomarker development and targeted therapeutic strategies across autonomic and post-viral disorders.

**Keywords** Postural Orthostatic Tachycardia Syndrome, Long COVID, Post-translational modifications, Proteomics, Coagulation

**Graphical abstract**




## Research insights

### What is known currently about this topic?

Long COVID and POTS are debilitating conditions with similar symptoms and symptom-based diagnoses. Previous proteomic studies focused on protein abundance, and the role of PTMs are unclear.

### What is the key research question?

Do PTMs in PPP-insoluble fraction distinguish pre-COVID-19 POTS, Long COVID, and Long COVID POTS?

### What is new?

PTMs distinguish PC-POTS, Long COVID, and LC-POTS.

Long COVID showed pro-coagulant PTMs; PC-POTS showed immune-metabolic PTMs.

PTM dysregulation explained severe symptoms despite normal routine blood tests.

### How might this study influence clinical practice?

PTM profiling may enable biomarker-guided diagnosis and targeted therapy.

## Background

Post Acute COVID-19 sequelae (PASC), or Long COVID, and Postural Orthostatic Tachycardia Syndrome (POTS) are chronic, debilitating, and primarily post-viral diseases [1]. Long COVID is characterized by persistent symptoms after acute COVID-19 infections, affecting 10–26% of cases [2, 3] across respiratory, systemic, musculoskeletal, neurological, cardiovascular, gastrointestinal, psychological, and dermatological categories [2, 4]. POTS

involves autonomic dysfunction and an abnormal heart rate increase (> 30 bpm) upon standing, predominantly in females (> 90%), often causing light-headedness or fainting [5–8].

Both Long COVID and POTS affect multiple physiological systems [9–11] but precise pathologies remain uncertain, and diagnoses are mostly symptom-based [12]. Despite severe symptoms, routine blood analyses often show normal protein levels, leading clinicians to treat symptoms rather than underlying causes.

In 2020, we identified dense amyloid-like deposits, called "microclots" in acute COVID-19 and Type 2 Diabetes Mellitus (T2DM) patients, implicating coagulation dysfunction [13]. In 2021, these deposits were observed in Long COVID blood, affecting the microcirculation [14]. We now refer to these microclots as fibrinaloid microclot complexes (FMCs) [15]. The insoluble fraction of platelet-poor plasma (PPP-IF) consists of FMCs, which are aggregates of inflammatory proteins and fibrinogen (present in both their normal and amyloid states), as well as neutrophil extracellular traps (NETs) and cellular debris. These cellular areas can be prothrombotic and can act as seeding areas for FMCs [15, 16]. Although healthy individuals may also contain FMCs within their PPP-IFs, the composition of PPP-IFs in patient groups may differ and could potentially explain severe symptoms [17] despite these pathological features being undetectable by conventional assays.

Following the COVID-19 pandemic, a positive correlation between Long COVID and POTS incidences has been observed [18–21] suggesting shared pathophysiology and raising the possibility that pre-COVID-19 POTS (PC-POTS) and Long COVID POTS (LC-POTS) may exhibit FMCs [22]. To our knowledge, the FMCs in PPP-IF of PC-POTS and LC-POTS have not yet been investigated, although it is known that many other diseases involving POTS and FMCs share comorbidities [22], and FMCs provide a mechanistically straightforward physiological explanation for POTS [22].

Previously, we conducted proteomics on FMCs in Long COVID to identify abundant proteins [23]. Systematic reviews indicate that potential biomarkers reflect multiple mechanisms, and no concise profile supports a diagnostic criterion [24]. Proteomics studies in POTS are limited, with mass-spectrometry analyses calling for further investigation of molecular mechanisms [12].

Given that conventional protein analyses fail to fully explain disease pathophysiology, we investigated post-translational modifications (PTMs) as a potential mechanism. PTMs, enzymatic or non-enzymatic, can be reversible or irreversible; oxidative and metabolic stresses often induce irreversible non-enzymatic PTMs [25]. PTMs on fibrinogen influence clot formation and degradation, contributing to pathological progression [26]. Previous studies of Long COVID PTMs focused only on soluble plasma proteins, thereby missing PTMs of proteins entrapped in the FMCs in the PPP-IF, potentially explaining severe symptoms despite normal lab values [27]. We note, too, that the proteome of FMCs differs substantially from that of normal clots, being highly enriched in amyloid(ogenic) proteins [28, 29]; the same is true for [30] the macroclots observed in [31] ischaemic stroke.

We thus hypothesized that by examining the PPP-IF, we would observe PTMs on proteins within FMCs which should reveal disease-specific molecular changes that could reflect pathology even when conventional protein levels appear normal. To test this, our cohort included healthy controls, PC-POTS, Long COVID with initial orthostatic hypotension (LC-IOH), and LC-POTS patients. The PPP-IF with FMCs was obtained by a double trypsin digest and LC-MS/MS proteomics. Analyses included protein profiling, biomarker discovery, and PTM evaluation.

Methods and materials

## Methods and materials

### Inclusion criteria for long COVID and healthy participants

Participants for Long COVID and healthy participants (controls) of ages between 18 and 80, male or female and who were able to sign and understand informed consent documentation were included. Participants with a positive SARS-CoV-2 test and who experience COVID-19 symptoms that lasted 12 weeks or longer after the confirmed infection were included in the Long COVID group. Participants who did not have a proven or suspected positive PCR SARS-CoV-2 test were included as controls. The status of their SARS-CoV-2 test was self-reported.

### Inclusion criteria for postural orthostatic tachycardia syndrome participants

Participants diagnosed with POTS within the age bracket of 18–60 years old, who were able to attend the Calgary Autonomic Research Clinic in Calgary and who are residents of Canada were included. Lastly, only participants whose sample were obtained before the COVID-19 pandemic were included in the PC-POTS group.

### Exclusion criteria for all participants

Participants were excluded if they could not provide informed consent, or if they could not safely withdraw from medicine use that could make test interpretation difficult or impossible. Other factors which in the investigator's opinion would prevent the participant from completing the protocol, including poor compliance during previous studies or an unpredictable schedule.

**Table 1** Demographics of participants. Categorical data are represented as percentages within groups and numerical data as the mean with standard deviation (SD) within groups

| Category | Healthy (*n* = 18) | LC-IOH (*n* = 19) | LC-POTS (*n* = 18) | PC-POTS (*n* = 20) |
|---|---|---|---|---|
| Female n (%) | 11 (61.1%) | 14 (73.7%) | 18 (100.0%) | 20 (100.0%) |
| Asian n (%) | 4 (22.2%) | 1 (5.3%) | 1 (5.6%) | 1 (5.0%) |
| Caucasian n (%) | 14 (77.8%) | 17 (89.5%) | 17 (94.4%) | 19 (95.0%) |
| Indigenous n (%) | 0 (0.0%) | 1 (5.3%) | 0 (0.0%) | 0 (0.0%) |
| COVID-19 Vaccinated n (%) | 18 (100.0%) | 19 (100.0%) | 18 (100.0%) | N/A (before COVID-19 pandemic) |
| Age (mean ± SD) | 47.33 ± 16.9 | 52.05 ± 11.08 | 37.11 ± 11.79 | 29.75 ± 7.61 |
| BMI (mean ± SD) | 26.48 ± 3.17 | 31.13 ± 7.98 | 24.16 ± 4.04 | 23.98 ± 5.49 |
| BSA (mean ± SD) | 1.92 ± 0.18 | 2.04 ± 0.3 | 1.77 ± 0.16 | 1.79 ± 0.23 |

*LC-IOH* Long COVID with initial orthostatic hypotension, *LC-POTS* Long COVID with Postural Orthostatic Tachycardia Syndrome, *PC-POTS* Pre-COVID-19 POTS, *BMI* Body Mass Index, *BSA* Body Surface Area

**Table 2** Multiple comparison test results for numerical demographic variables. A completely randomized analysis of variance was performed for age, BMI and BSA followed by a Tukey post-hoc test. Where residuals were not normally distributed a non-parametric Kruskal-Wallis test was performed followed by a post-hoc Dunn's test with a Bonferroni correction. In the case of a heteroscedastic distribution of data over groups, the data were log transformed

| Comparison | Age Tukey's HSD *p*-value (data log transformed due to heteroscedasticity) | BMI Dunn's Test *p*-value | BSA Tukey's HSD *p*-value |
|---|---|---|---|
| Healthy vs. LC-POTS | 0.1330 | 0.4954 | 0.2095 |
| Healthy vs. PC-POTS | 0.0004* | 0.1523 | 0.3351 |
| Healthy vs. LC-IOH | 0.5116 | 1 | 0.3502 |
| LC-POTS vs. PC-POTS | 0.1970 | 1 | 0.9878 |
| LC-POTS vs. LC-IOH | 0.0030* | 0.0109* | 0.0027* |
| PC-POTS vs. LC-IOH | 0.0000* | 0.0015* | |

*BMI* Body Mass Index, *BSA* Body Surface Area

*$p < 0.05$

### Diagnostic criteria

#### *Long COVID sub-groups*

Participants who met the inclusion criteria for Long COVID were further sub-divided into Long COVID patients with initial orthostatic hypotension (LC-IOH) or Long COVID with Postural Orthostatic Tachycardia Syndrome (LC-POTS). LC-IOH was diagnosed when a transient drop in SBP of > 40mmHg within 15s of standing, followed by recovery within 60s of standing, was observed as described in the "Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome" [32]. LC-POTS were diagnosed when a HR increase > 30 bpm within 5-10 min of standing in addition to chronic symptoms were observed.

#### *POTS*

Physician diagnosis of POTS as defined by the Canadian Cardiovascular Society Consensus Statement [33] and sustained, excessive orthostatic tachycardia ≥ 30 bpm within 10 min standing, absence of orthostatic hypotension (> 20/10 mmHg decrease) within 3 min standing and chronic orthostatic symptoms that improve with recumbence.

### Study Participant details

Table 1 summarises demographic details of all included participants.

Table 2 summarises the demographic differences between disease groups of the cohort. A higher proportion of female participants in the POTS groups was expected, as this condition predominantly affects females. The 100% vaccination rate observed across all post–COVID-19 groups minimize the potential confounding effects of vaccination on the results.

Age differences were statistically significant between the PC-POTS group and all other groups, which may be attributable to differences in the inclusion age ranges.

Body mass index (BMI) and body surface area (BSA) were significantly higher in the LC-IOH group compared to both the LC-POTS and PC-POTS groups. This should be taken into account when interpreting differences in metabolism-related proteins between disease groups. However, as no statistically significant differences were observed between disease groups and healthy controls, meaningful inferences regarding metabolic pathways remain valid.

Within the LC-POTS group, one participant had mast cell activation syndrome and small intestinal bacterial overgrowth, another presented with Post-exertional Malaise (PEM) and chronic fatigue syndrome (CFS), and a third had a diagnosis of depression. In the LC-IOH group, two participants had diabetes, one had Sjögren's disease and lichen planus, and another had hypothyroidism. All reported comorbidities were diagnosed prior to the onset of Long COVID or POTS. The medicinal use of the cohort is available in the supplementary data in Table S1.

### Ethics and informed consent

Human samples used in this research were obtained from the University of Calgary where they had been collected under approvals from The Conjoint Health Research Ethics Board (CHREB), University of Calgary (REB21-1188 and REB15-2434).

The transfer of samples from the University of Calgary to the University of Stellenbosch was conducted under

a formal Material Transfer Agreement (MTA) (Contract number: S009247). The MTA governed the conditions of sample transfer, use, storage, and confidentiality, and confirmed that all materials were provided in accordance with the original ethics approval. All samples were received in de-identified form and were used solely for the purposes described in this study. No new samples were collected specifically for this study.

#### Sample collection

Samples were drawn by phlebotomists and were centrifuge within 30 min of the blood draw at 3000 rpm for 10 min at 4 °C and frozen at -80 °C within 30 min after centrifugation. Samples underwent 2 freeze-thaw cycles i.e. samples were thawed for a second time when proteomics experiments were conducted.

#### Sample preparation

A 19× dilution of platelet-poor plasma (PPP) was prepared using 10 mM ammonium bicarbonate (AmBic). AmBic is a volatile buffer that does not form crystals during the drying process, which is often observed with phosphate-buffered saline (PBS), and it functions at the optimal pH – 7.8–8.2. The solution was then centrifuged at 12,000 × g for 10 min. The supernatant was removed, and the remaining sample was subjected to a double trypsin digestion.

Firstly, the more soluble fraction was digested using a low concentration of trypsin at a ratio of 1:5000 protease: protein concentration (final trypsin concentration 0.574 ng/µL). This was followed by overnight incubation at room temperature. The next day, samples were centrifuged at 12,000 × g for 20 min.

This was followed by a reduction step. The supernatant (92 µL) was removed, and the remaining 8 µL along with the pellet was incubated with ~5 mM tris(2-carboxyethyl) phosphine (TCEP) in 200 mM triethylammonium bicarbonate (TEAB) (denaturing buffer; pH 8). The samples were incubated in a heating block at 60 °C for 45 min, after which they were cooled to room temperature. S-methyl methanethiosulfonate (MMTS) was added to a final concentration of 10 mM to block free cysteine residues and prevent disulfide bond reformation.

The second digestion was performed at a 1:50 protease: protein ratio (final trypsin concentration of 6 ng/µL), in 40mM TEAB with 10% methanol (MeOH) as a denaturing solvent. This mixture was incubated for 18 h at 37 °C. To precipitate the trypsin, trifluoroacetic acid (TFA) was added to a final concentration of 1%. The mixture was briefly vortexed and then centrifuged at 16,000 × g for 15 min. The supernatant was removed, dried, and used for peptide quantification as a standardization step prior to liquid chromatography–tandem mass spectrometry (LC-MS/MS).

#### Sample analysis on the EvoSep TimsTOF Pro-2

Samples were analyzed by data-independent acquisition liquid chromatography tandem mass spectrometry (DIA LC-MS/MS) using an EvoSep One LC system coupled to a timsTOF Pro 2 mass spectrometer equipped with trapped ion mobility spectrometry (TIMS). Samples were diluted in 5% formic acid and loaded onto EvoTip Pure tips (Evosep ApS, EV2011) according to the manufacturer's instructions, then stored in 0.1% formic acid until analysis. Chromatographic separation was performed using standard EvoSep gradients of 100, 60, 30, and 15 samples per day (SPD). Mobile phases consisted of 0.1% formic acid in water (A) and 0.1% formic acid in acetonitrile (B). Separation was performed using EvoSep Performance columns (8 cm × 150 µm, 1.5 µm C18 packing) for the 100 and 60 SPD gradients and EvoSep Performance columns (15 cm × 150 µm, 1.5 µm C18 packing) for the 30 and 15 SPD gradients. The columns were coupled to a Bruker timsTOF Pro 2 via a 20 µm silica emitter (Bruker CaptiveSpray ZDV Sprayer 20) and maintained at 40 °C using a Bruker Column Toaster. Mass spectrometry data were acquired in diaPASEF mode. Raw Bruker files were processed using FragPipe v24.0 incorporating MSFragger v4.0, diaTracer v2.2.1, and DIA-NN v2.0.2.

#### Statistical analyses

A spectral library was generated during the Fragpipe analysis as part of the DIANN workflow with 1% false discovery rate (FDR) cutoff. This library.tsv file was imported into Skyline. The database search was performed with a precursor charge range of 1–4, while fragment ions were considered at a maximum charge state of 1. Trypsin was specified as the proteolytic enzyme, allowing up to two missed cleavages per peptide. The precursor mass tolerance was set to ± 15 ppm for the initial search, with a refined (true) tolerance of ± 20 ppm, and the fragment ion mass tolerance was set to ± 20 ppm. Spectral matching was conducted using b- and y-type fragment ions, requiring a minimum of four matched fragment ions for peptide-spectrum match (PSM) reporting, with at least two matched fragments required for inclusion in statistical modelling. For spectral preprocessing, the top 150 most intense peaks per spectrum were retained, with a minimum requirement of 15 peaks per spectrum and an intensity threshold set at 1% of the base peak.

Group comparisons were performed within Skyline, and adjusted p-values (Benjamini–Hochberg correction for multiple comparisons) as well as fold changes were extracted. Statistically significant peptides (adjusted p-value < 0.05) identified in the initial analysis were used to generate a list of proteins. All peptides of the selected proteins were subsequently validated by confirming that at least four fragment ions co-eluted within the same retention time window. Following validation,

**Table 3** List of proteins used for final protein, peptide and post-translational modifications analysis. The protein list was added to Metascape software and Gene Ontology was used to manually classify each protein under the broader categories Coagulation, Immunity, and Metabolic. Gene ontology as obtained through Metascape was added to the table for each protein

| Proteins | Description | Gene Ontology Biological Process as Determined By Protein Enrichment Through Metascape |
|---|---|---|
| COAGULATION | | |
| ANTIII | antithrombin III | GO:0030193 regulation of blood coagulation; GO:1,900,046 regulation of hemostasis; GO:0050818 regulation of coagulation |
| FGA | fibrinogen alpha chain | GO:2,000,258 negative regulation of protein activation cascade; GO:2,000,260 regulation of blood coagulation, common pathway; GO:2,000,261 negative regulation of blood coagulation, common pathway |
| FGB | fibrinogen beta chain | GO:0043152 induction of bacterial agglutination; GO:0031639 plasminogen activation; GO:0034116 positive regulation of heterotypic cell-cell adhesion |
| FGG | fibrinogen gamma chain | GO:0031639 plasminogen activation; GO:0034116 positive regulation of heterotypic cell-cell adhesion; GO:0042730 fibrinolysis |
| ITGA2B | integrin subunit alpha 2b | GO:0033631 cell-cell adhesion mediated by integrin; GO:0033627 cell adhesion mediated by integrin; GO:0034113 heterotypic cell-cell adhesion |
| MYH9 | myosin heavy chain 9 | GO:0032796 uropod organization; GO:0060471 cortical granule exocytosis; GO:1,903,918 regulation of actin filament severing |
| IMMUNITY | | |
| A2M | alpha-2-macroglobulin | GO:0001868 regulation of complement activation, lectin pathway; GO:0001869 negative regulation of complement activation, lectin pathway; GO:0045916 negative regulation of complement activation |
| C3 | complement C3 | GO:0001970 positive regulation of activation of membrane attack complex; GO:0097278 complement-dependent cytotoxicity; GO:0150064 vertebrate eye-specific patterning |
| C4A | complement C4A (Chido/Rodgers blood group) | GO:2,000,427 positive regulation of apoptotic cell clearance; GO:2,000,425 regulation of apoptotic cell clearance; GO:0008228 opsonization |
| C4B | complement C4B (Chido/Rodgers blood group) | GO:2,000,427 positive regulation of apoptotic cell clearance; GO:2,000,425 regulation of apoptotic cell clearance; GO:0008228 opsonization |
| IC1 | Inhibitor of Complement factor 1 | GO:0001868 regulation of complement activation, lectin pathway; GO:0001869 negative regulation of complement activation, lectin pathway; GO:0045916 negative regulation of complement activation |
| ITIH4 | inter-alpha-trypsin inhibitor heavy chain 4 | GO:0030212 hyaluronan metabolic process; GO:0006953 acute-phase response; GO:0030203 glycosaminoglycan metabolic process |
| IGHG1 | immunoglobulin heavy constant gamma 1 (G1m marker) | GO:0097278 complement-dependent cytotoxicity; GO:0001788 antibody-dependent cellular cytotoxicity; GO:0001794 type IIa hypersensitivity |
| METABOLIC | | |
| APOA1 | apolipoprotein A1 | GO:0018206 peptidyl-methionine modification; GO:1,902,994 regulation of phospholipid efflux; GO:1,902,995 positive regulation of phospholipid efflux |
| APOB | apolipoprotein B | GO:0006642 triglyceride mobilization; GO:0010886 positive regulation of cholesterol storage; GO:0090077 foam cell differentiation |
| ALB | albumin | GO:0015723 bilirubin transport; GO:0072732 cellular response to calcium ion starvation; GO:0051902 negative regulation of mitochondrial depolarization |
| HBA1 | hemoglobin subunit alpha 1 | GO:0030185 nitric oxide transport; GO:0015670 carbon dioxide transport; GO:0015671 oxygen transport |
| HBB | hemoglobin subunit beta | GO:0030185 nitric oxide transport; GO:0015670 carbon dioxide transport; GO:0015671 oxygen transport |
| ZA2G | alpha-2-glycoprotein 1, zinc-binding | GO:0002486 antigen processing and presentation of endogenous peptide antigen via MHC class I via ER pathway, TAP-independent; GO:0002476 antigen processing and presentation of endogenous peptide antigen via MHC class Ib; GO:0002484 antigen processing and presentation of endogenous peptide antigen via MHC class I via ER pathway |

protein- and peptide-level comparisons were conducted for the established protein list. Fold changes between groups were calculated, and $\log_2$ fold change ($\log_2$FC) values were derived. Adjusted p-values were transformed to $-\log_{10}$(p-value) for volcano plot visualization. Significance thresholds were set at $|\log_2 FC| > 1$ and adjusted p-value < 0.05, with features exceeding both thresholds classified as significantly differentially abundant. All statistical analyses were performed using Python 3.14.2 with the SciPy library. Data visualization was conducted using the Matplotlib and Seaborn packages.

## Results

A list of proteins of interest (Table 3) was generated by identifying the statistically significant peptides (adjusted p-value < 0.05) of an initial analysis. Subsequently, all

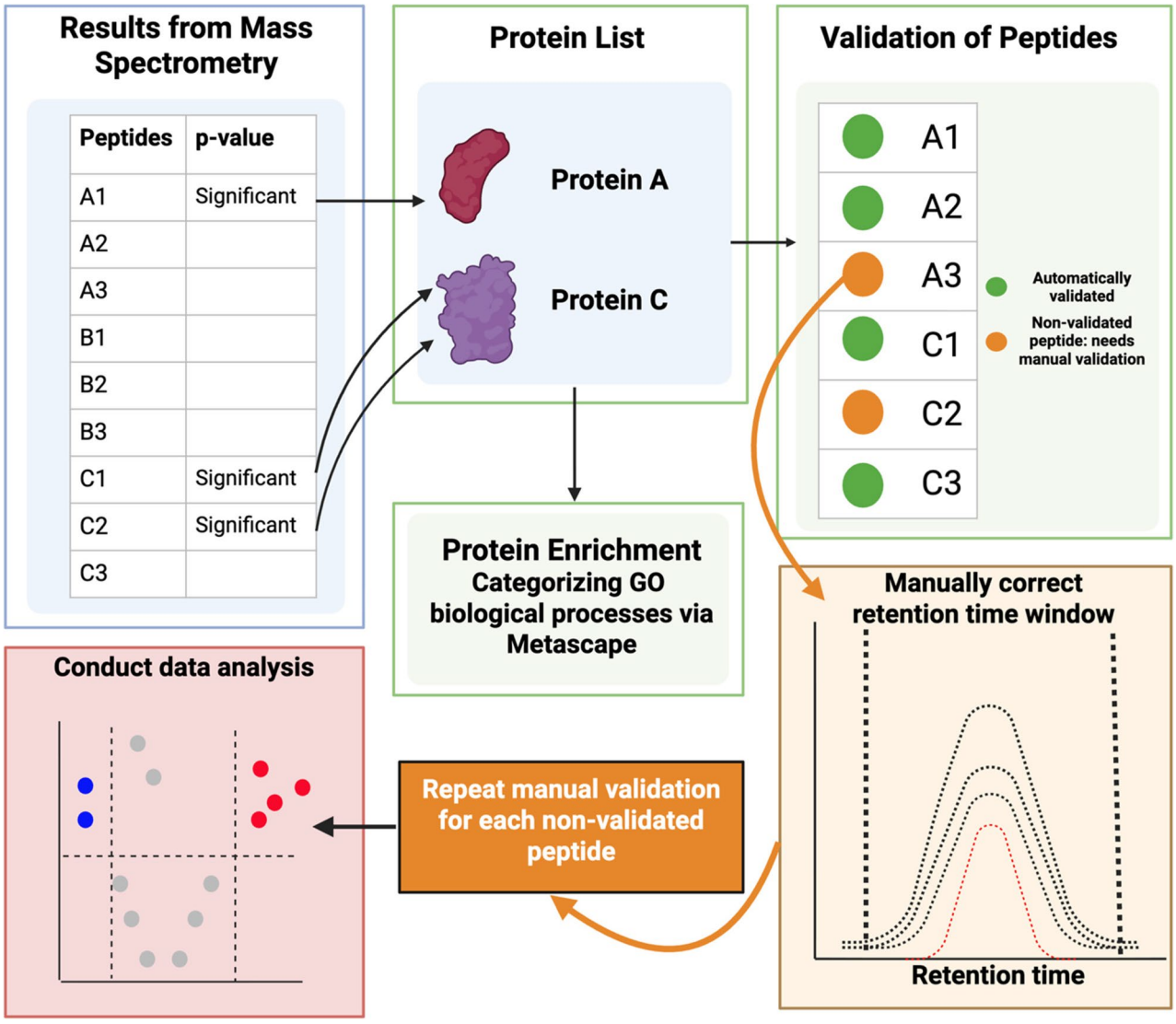



**Fig. 1** Protein selection process. This figure illustrates the filtering and selection workflow that was used to identify proteins of interest. It outlines how proteins were filtered and systematically refined into a dataset ready for downstream analysis. Furthermore, it highlights the use of protein enrichment analysis using Metascape to establish gene ontology (GO) biological processes. From the mass-spectrometry results, statistically significant peptides (from controls vs. disease group) were identified and the protein list of interest was established. This list was used for protein enrichment, and each peptide of the selected proteins were validated if not already automatically validated. Validation was achieved by manually adjusting the retention time window. This step was repeated for each peptide. There after data analyses could be conducted. This diagram was created in BioRender

peptides of these proteins were validated before further data extraction.

Given the complexity of the proteomic dataset, multiple filtering steps were applied prior to downstream analysis (Fig. 1). In Skyline, peptide identifications are annotated based on integration and spectral matching quality. High-confidence peptides were retained for quantitative analysis; however, a subset of peptides with only four matched fragment ions within the retention time window were flagged for further evaluation (orange).

An initial screening analysis was then performed to identify statistically significant dysregulated peptides between disease and control groups. The corresponding proteins were compiled to generate a list of proteins of interest. Within this subset, peptides flagged in orange underwent manual validation, requiring confirmation that at least all four fragment ions' peaks were present within the appropriate retention time window to ensure robust quantification.

Functional enrichment analysis of the resulting protein set was performed using Metascape [34], identifying associated Gene Ontology (GO) biological processes. The GO biological processes were used to manually and broadly categorize the proteins under coagulation, immunity and metabolic pathways (Table 3).

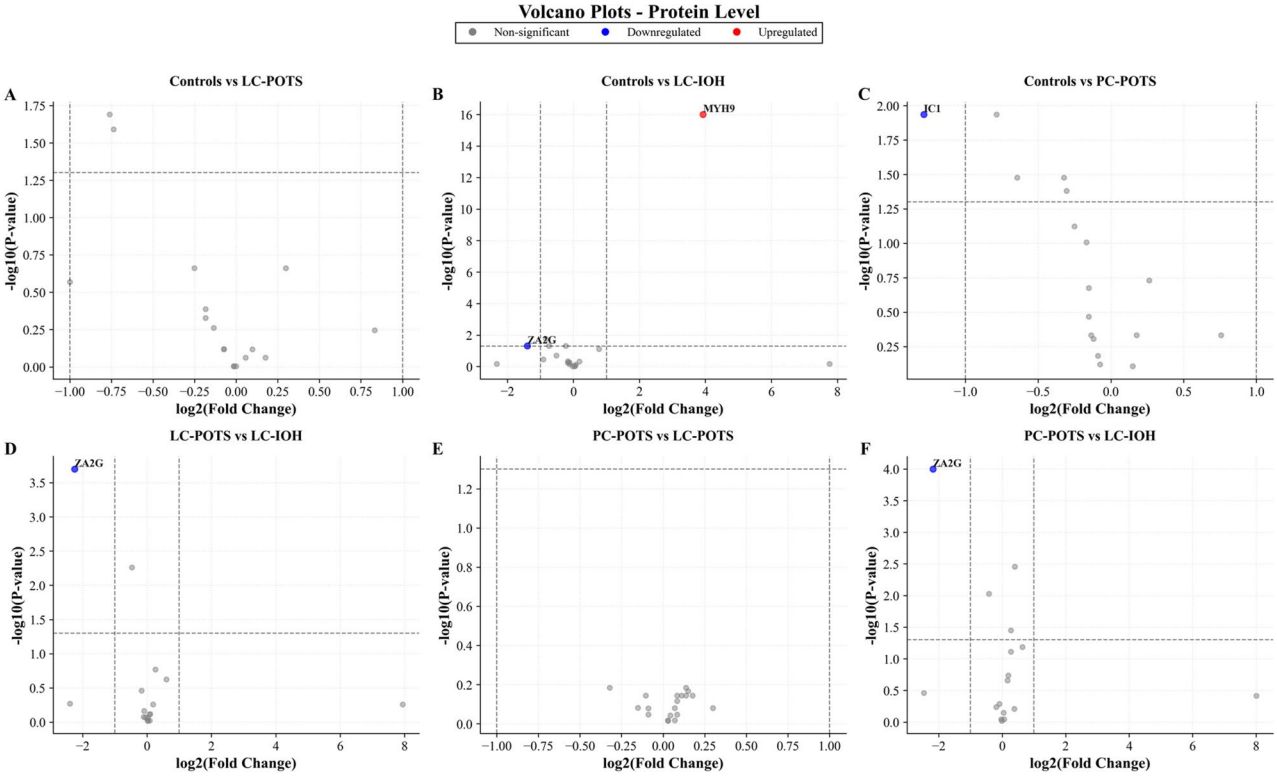



**Fig. 2** Volcano plots illustrating differential protein regulation across pairwise group comparisons. For protein-level comparisons, fold changes were calculated between groups, and log2 fold change (log2FC) values were computed. Adjusted p-values (corrected with Benjamin-Hochberg for multiple group comparison) were transformed to -log10(p-value) for visualization. The x-axis represents log2FC, and the y-axis represents -log10(adjusted p-value). Significance thresholds were set at $|log2FC| > 1$ and adjusted p-value < 0.05. Features exceeding both thresholds were classified as significantly differentially abundant. Horizontal dotted lines indicate the -log10(p-value) cutoff, and vertical dotted lines indicate the log2 fold change cutoff (± 1). **A** Controls vs. LC-POTS: No proteins were significantly different. **B** Controls vs. LC-IOH: Two proteins were significantly different (MYH9 upregulated; ZA2G downregulated). **C** Controls vs. PC-POTS: Only IC1 was significantly downregulated. **D** LC-POTS vs. LC-IOH: ZA2G was significantly downregulated. **E** PC-POTS vs. LC-POTS: No proteins were significantly different. **F** PC-POTS vs. LC-IOH: ZA2G was significantly downregulated. POTS: Postural Orthostatic Tachycardia Syndrome, PC: Pre-COVID; LC: Long COVID; IOH: Initial Orthostatic Hypotension

### Proteomic analysis of the PPP-IF revealed minimal protein differences

Proteins that were dysregulated in at least one disease group relative to controls were included in comparisons between disease groups. Only three proteins showed dysregulation at the protein concentration level (Fig. 2). ZA2G was downregulated and MYH9 was upregulated in LC-IOH compared to controls. Additionally, ZA2G was downregulated in LC-IOH when compared to PC-POTS and LC-POTS (i.e. ZA2G is higher in PC-POTS and LC-POTS than in LC-IOH). IC1 was downregulated in PC-POTS relative to controls. Detailed results are available in Supplementary Table S2.

### There is extensive PTM dysregulation within the PPP-IF

The proteomics analysis which included PTMs revealed 29 unique base peptides which showed significant differences between controls and at least 1 disease group and 58.62% of the base peptides were amyloidogenic according to Amylogram (results in Supplementary Table S3.1 and S3.2), i.e. had AmyloGram scores greater than 0.5. Furthermore, these 29 base peptides originating from 16 proteins, included 64 uniquely modified peptides and revealed multiple significant differences between groups (Fig. 3; Table 4). Detailed results can be found in Supplementary Tables S4.

## Discussion

In this study, we investigated whether the peptides with PTM of proteins within the PPP-IF distinguish PC-POTS, LC-IOH and LC-POTS at the molecular level. The PC-POTS samples were collected before the COVID-19 pandemic, enabling us to define the baseline molecular signatures of classic POTS entirely independent of SARS-CoV-2 exposure. The LC-POTS group were all diagnosed with POTS after an acute COVID-19 infection. The pre-pandemic POTS cohort therefore provides a true reference for classical POTS biology, allowing us to distinguish molecular features that originate from established POTS mechanisms from those newly introduced or amplified by Long COVID in individuals who later developed POTS symptoms. The composition of

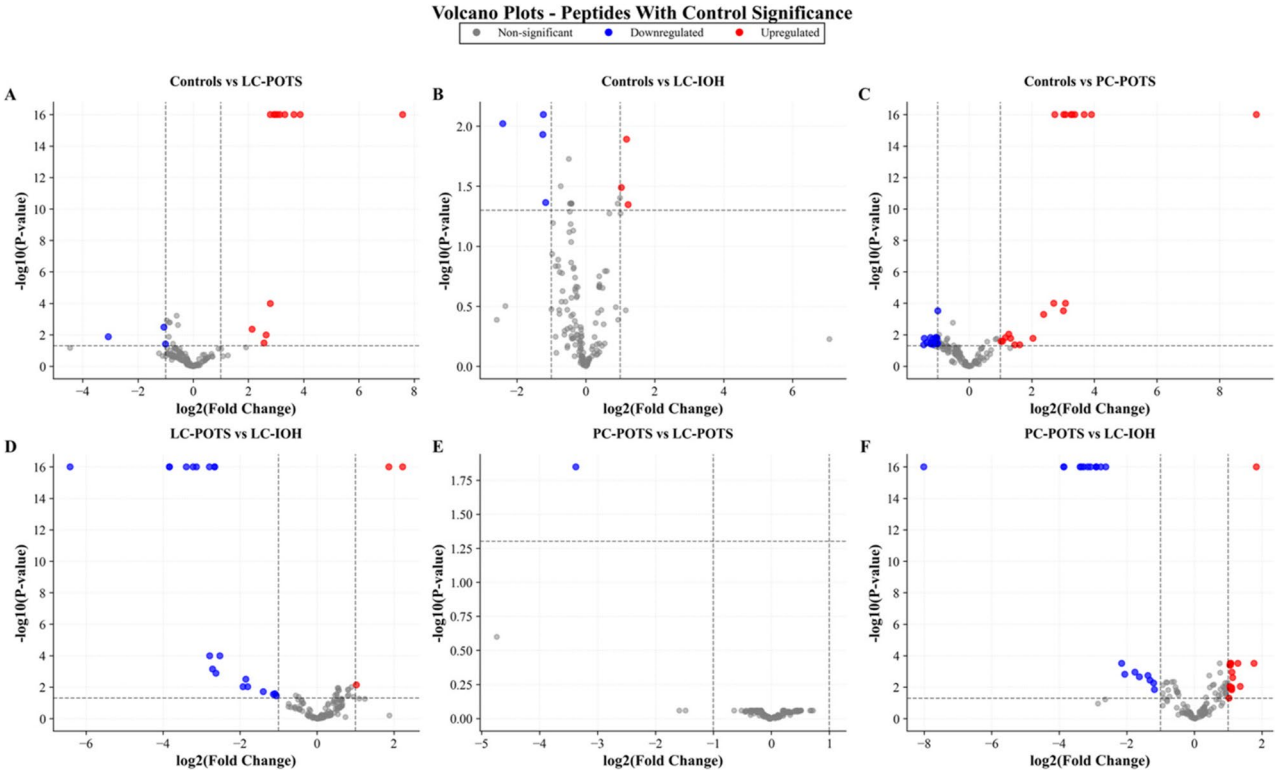



**Fig. 3** Volcano plots illustrating differential peptides with Post-Translational Modifications (PTMs) across pairwise group comparisons. For peptide-level comparisons, pre-calculated log2 fold change values from mass spectrometry data were used. Adjusted p-values were transformed to -log10(p-value) for visualization. The x-axis represents log2 (fold change), and the y-axis represents -log10(p-value). Significance thresholds were set at |log2FC| > 1 and adjusted p-value < 0.05. Features exceeding both thresholds were classified as significantly differentially abundant. Horizontal dotted lines indicate the -log10(p-value) cutoff, and vertical dotted lines indicate the log2 fold change cutoff (± 1). Analysis of 29 unique base peptides and 64 uniquely modified peptides across 16 proteins revealed several significant differences between groups. **A** Controls vs. LC-POTS: 15 uniquely modified peptides (8 base peptides from 6 proteins) were dysregulated (12 upregulated, 3 downregulated). **B** Controls vs. LC-IOH: 7 uniquely modified peptides (7 base peptides from 6 proteins) were dysregulated (3 upregulated, 4 downregulated). **C** Controls vs. PC-POTS: 44 uniquely modified peptides and one unmodified peptide (22 base peptides from 12 proteins) were dysregulated (20 upregulated, 25 downregulated). **D** LC-POTS vs. LC-IOH: 22 uniquely modified peptides (11 base peptides from 8 proteins) were significantly different (3 upregulated, 19 downregulated). **E** PC-POTS vs. LC-POTS: 1 downregulated modified peptide. **F** PC-POTS vs. LC-IOH: 41 uniquely modified peptides and one unmodified peptide (14 base peptides from 8 proteins) were dysregulated (22 upregulated, 20 downregulated). POTS: Postural Orthostatic Tachycardia Syndrome, PC: Pre-COVID; LC: Long COVID; IOH: Initial Orthostatic Hypotension

the PPP-IF was achieved with proteomic analysis and revealed that widespread dysregulation of PTMs far exceeded changes observed at the level of protein abundance. This suggests that PTMs, rather than protein concentration, more accurately reflect the underlying disease biology.

### Coagulation-related PTMs and FMC structural biology

Several fibrinogen peptides displayed disease-associated PTM patterns, underscoring the central role of coagulation proteins in PPP-IF composition. In LC-IOH, increased advanced glycation end-products (AGEs) modifications on FIBA peptides align with known glycation patterns observed in diabetes [35]. Furthermore, AGEs result in the formation of covalent cross-links, which may help explain the robust, fibrinolysis-resistant nature of FMCs within PPP-IF in this group [36]. Conversely, PC-POTS showed reduced modification of selected FIBG peptides, which may reflect structural shielding of these sites once fibrinogen is incorporated into compact FMCs [37] or it may be the age differences between the cohorts with the PC-POTS group being significantly younger than the controls. Oxidation, deamination and pyroglutamate (PYD) formation of fibrinogen subunits could collectively suggest altered clot architecture across diseases, with implications for fibrinolysis, clot permeability, and microvascular obstruction, which was also seen in the LC-POTS group. The presence of amyloidogenic peptides in multiple fibrinogen regions supports the concept that FMCs arise from misfolded, β-sheet-rich aggregates rather than classical thrombin-driven fibrin [38].

At the protein level, MYH9 is upregulated in LC-IOH alongside peptides of ITA2B. These results are consistent with PTM changes indicative of dysregulated anticoagulant function and could allude to altered platelet–fibrinogen interactions [39, 40]. The latter could also apply to

**Table 4** Results from pair wise peptide and PTM comparison. For all comparisons (X vs Y), regulation is defined relative to the second group (Y). Up-regulation and down-regulation therefore reflect increased or decreased levels in Y compared with X, respectively

| Comparison | TotalProteins** | Total Base Peptide Sequences* | Total dys-regulated modified peptides | Total dys-regulated un-modified Peptides | Total up regulated*** | Total down-regulated*** |
|---|---|---|---|---|---|---|
| Coagulation | | | | | | |
| Controls vs LC-IOH | 2 | 2 | 2 | 0 | 2 | 0 |
| Controls vs LC-POTS | 1 | 1 | 1 | 0 | 1 | 0 |
| Controls vs PC-POTS | 3 | 4 | 16 | 0 | 2 | 14 |
| LC-POTS vs LC-IOH | 3 | 3 | 3 | 0 | 1 | 2 |
| PC-POTS vs LC-IOH | 3 | 4 | 21 | 0 | 19 | 2 |
| PC-POTS vs LC-POTS | 0 | 0 | 0 | 0 | 0 | 0 |
| Immunity | | | | | | |
| Controls vs LC-IOH | 1 | 2 | 2 | 0 | 1 | 1 |
| Controls vs LC-POTS | 3 | 5 | 9 | 0 | 7 | 2 |
| Controls vs PC-POTS | 6 | 12 | 20 | 0 | 13 | 7 |
| LC-POTS vs LC-IOH | 3 | 6 | 14 | 0 | 2 | 12 |
| PC-POTS vs LC-IOH | 3 | 6 | 14 | 0 | 1 | 13 |
| PC-POTS vs LC-POTS | 1 | 1 | 1 | 0 | 0 | 1 |
| Metabolic | | | | | | |
| Controls vs LC-IOH | 3 | 3 | 3 | 0 | 0 | 3 |
| Controls vs LC-POTS | 2 | 2 | 5 | 0 | 4 | 1 |
| Controls vs PC-POTS | 3 | 6 | 8 | 1 | 5 | 4 |
| LC-POTS vs LC-IOH | 2 | 2 | 5 | 0 | 0 | 5 |
| PC-POTS vs LC-IOH | 2 | 4 | 6 | 1 | 2 | 5 |
| PC-POTS vs LC-POTS | 0 | 0 | 0 | 0 | 0 | 0 |

*Total base peptides are those peptides that were modified and which's modified, or unmodified, forms showed differences between groups. **Total proteins are the proteins represented by these base peptides. ***Total up- and downregulated counts refer to both the modified and unmodified forms of peptides that differed significantly between groups. All data that was used to generate the results in this table are reported in detail in the supplementary tables S2 and S4

the PC-POTS group as dysregulation of ANT3 peptides was observed within that group. Collectively, these findings possibly indicate that the coagulation system is particularly disrupted in LC-IOH and LC-POTS (Fig. 4), with FMC-associated PTMs potentially amplifying microvascular dysfunction.

#### Immune and complement system dysregulation

PTMs in proteins involved in complement activation and innate immunity has the potential to further differentiate the disease groups. C3, C4A/B and IGG1 peptides displayed distinct PTM profiles in PC-POTS compared to controls, suggesting altered complement turnover and immune activation [41]. Downregulation of IC1 in PC-POTS parallels patterns seen in complement/contact activation disorders such as hereditary angioedema, which share clinical overlap with PC-POTS, including hypotension, autonomic instability and inflammatory symptoms. Additionally, the complement system plays a central role in mast cell activation and allergic inflammation [42], and several clinical studies have reported substantial overlap between mast cell activation syndrome (MCAS) and POTS [43].

A2MG showed multiple oxidation- and deamination-based PTMs, particularly in PC-POTS and LC-POTS. Given A2MG's role in trapping proteases and binding cytokines, these modifications could alter inflammatory signalling [44, 45]. The pattern observed for ITIH4 (reduced oxidised forms in PC-POTS despite its identity as an acute-phase reactant) may indicate an oxidative environment that selectively preserves functional ITIH4, potentially contributing to sustained inflammatory signalling [46]. Collectively, these findings could suggest distinct immune-modulatory landscapes across diseases, especially in PC-POTS and LC-POTS groups.

#### Metabolic pathway dysregulation in PC-POTS and LC-POTS

PTMs in lipid-handling proteins were prominent in PC-POTS and LC-POTS. Oxidised and deaminated variants of apoA1 and apoB were consistently upregulated in PC-POTS relative to both controls and LC-IOH. This pattern resembles PTM signatures observed in T2DM, where oxidised apolipoproteins contribute to vascular dysfunction and altered lipid metabolism [47, 48]. ZA2G downregulation in PC-POTS and LC-IOH aligns with insulin resistance pathways and potentially supports a metabolic contribution to symptoms [49, 50]. These findings could indicate that PC-POTS is characterized by a pronounced metabolic-oxidative PTM profile. Moreover, because the metabolic dysregulation observed in LC-IOH - specifically AGE- and oxidation-mediated modifications of fibrinogen - mirrors that seen in T2DM, the LC-IOH group

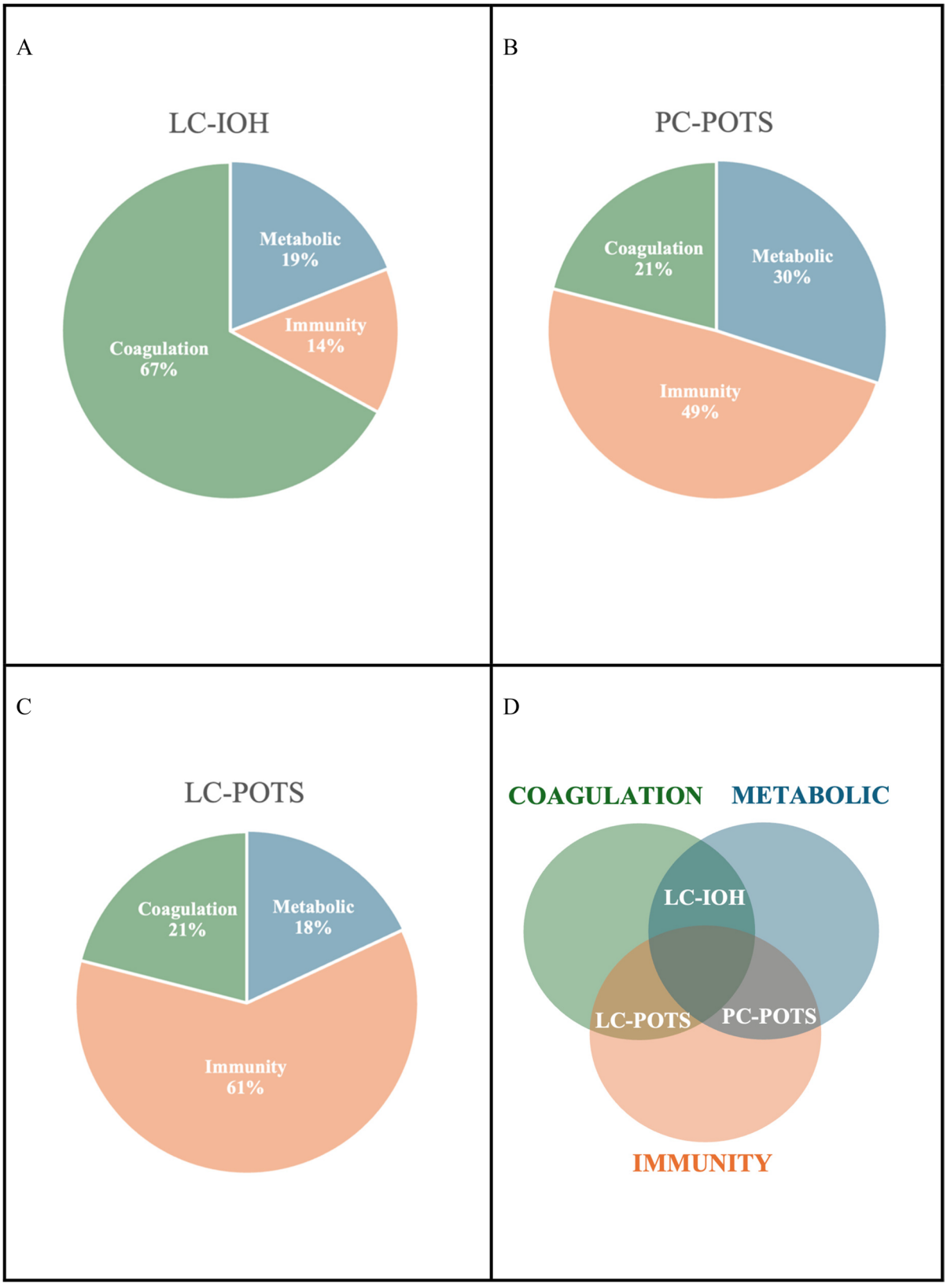



**Fig. 4** (See legend on next page.)

(See figure on previous page.)

**Fig. 4** Visualisation of a hypothesis towards functional category contribution of dysregulated peptides across disease groups based on PTM and protein level dysregulation. **A**-**C**. Pie charts illustrate the relative contribution of dysregulated peptides and proteins to three functional categories (metabolic, coagulation, and immunity) within each disease group. For each group, $log_2$ fold-change values were calculated relative to controls and from pairwise comparisons between disease groups, and peptides were included according to the direction of regulation consistent with the expected biological effect of disease on the respective pathways. $Log_2$ fold-change values of dysregulated peptides and proteins were summed within each functional category to generate a relative measure of category contribution, which was then expressed as a percentage of the total summed signal per disease group. These values are intended to provide a visual summary of dominant biological themes and were not used for statistical inference. **D** A schematic Venn diagram summarizes the dominant functional categories across disease groups based on the two highest-contributing categories per group, highlighting conceptual overlap in biological processes rather than shared peptides or proteins.LC-IOH: Long COVID with initial orthostatic hypotension, LC-POTS Long COVID with Postural Orthostatic Tachycardia Syndrome; PC-POTS: Pre-COVID 19 POTS

likewise exhibits dysregulation of metabolic pathways (Fig. 4). Although two participants had pre-existing type 2 diabetes, these individuals were retained in the analysis as diabetes represents a biologically relevant metabolic comorbidity that may act as a predisposing or interacting factor in post-acute disease states. Exclusion of such cases could therefore introduce bias by removing clinically meaningful heterogeneity inherent to the LC-IOH cohort. Importantly, the observed metabolic signature was consistent across the group and was not driven by these individuals. This was confirmed with a repeat of the analysis where they were indeed removed from the cohort.

#### Cross-disease integration and mechanistic interpretation

Synthesising these data, LC-IOH appears to be predominantly characterised by PTM dysregulation of coagulation proteins, particularly fibrinogen, with downstream effects on both coagulation and metabolic pathways. In contrast, both PC-POTS and LC-POTS exhibit more pronounced PTM alterations within immune pathways; PC-POTS shows secondary involvement of metabolic modifications, whereas LC-POTS demonstrates secondary dysregulation of coagulation pathways. The increase in AGE-modified FIBA peptides in LC-IOH mirrors glycation patterns in diabetes; a convergence with notable clinical relevance, given that metformin reduces the risk of developing Long COVID [51]. The downregulation of FIBG modifications in PC-POTS suggests structural differences in FMC architecture that may shield specific fibrinogen sites from modification, although the age difference between the cohorts could also be contributing factor. The high amyloidogenicity of many dysregulated peptides further reinforces the nature of FMCs as aggregated, amyloid-prone complexes. Importantly, the differential modification of individual peptides within the same protein indicates site-specific PTM non-enzymatic modifications rather than uniform protein-level changes, offering a potential molecular explanation for why routine soluble plasma assays often fail to capture disease pathology.

#### Clinical relevance and implications for biomarker development

Samples from LC-IOH patients exhibited widespread PTM alterations in coagulation-related proteins, particularly AGE- and oxidation-based modifications of fibrinogen subunits, suggesting a strong pro-coagulant signature, whereas PC-POTS presented a predominantly metabolic-oxidative PTM pattern affecting apolipoproteins and pointing to metabolic dysregulation. Secondarily, in PC-POTS it affects the immune system through IGG1 and complement systems proteins. Interestingly, both conditions share features commonly associated with diabetes, such as protein glycation and oxidative stress. LC-POTS shares features with LC-IOH in the coagulation pathway, and with PC-POTS in immunity (Fig. 4). Together, these findings highlight PTMs within the PPP-IF as a mechanistically rich and clinically relevant domain of investigation. By analysing the PPP-IF, we have uncovered biochemical perturbations invisible to standard blood tests, offering a potential explanation for persistent and debilitating symptoms despite "normal" laboratory values. PTM signatures, especially those involving fibrinogen, complement proteins, and apolipoproteins, may hold utility as diagnostic or stratification biomarkers, and the distinct PPP-IF associated PTM patterns observed across LC-IOH, PC-POTS and LC-POTS could inform disease-specific therapeutic strategies, including interventions targeting glycation pathways, oxidative stress, or complement activation.

#### Limitations and future directions

The PPP samples underwent two freeze-thaw cycles which could induce PTM's and protein degradation, however this limitation was shared equally by all samples which was subjected to identical conditions. This study focused exclusively on the PPP-IF; parallel analysis of soluble proteins would clarify how PTMs partition between fractions. Larger cohorts are needed to validate disease-specific PTM signatures and to correlate molecular findings with symptom severity. Furthermore, the entire cohort are from Canadian residents who visited a single clinic and therefore the generalization of the findings might be limited. The use of Amylogram remains an in silico predictive tool and the results obtain from the model should be used with caution, it still provides a useful framework and potentially aid in disease and FMC characterisation. Functional assays are required to determine how individual PTMs affect protein activity, fibrinogen structure, or complement regulation. Including a T2DM cohort in future studies will be essential, given the strong overlap between LC-IOH and

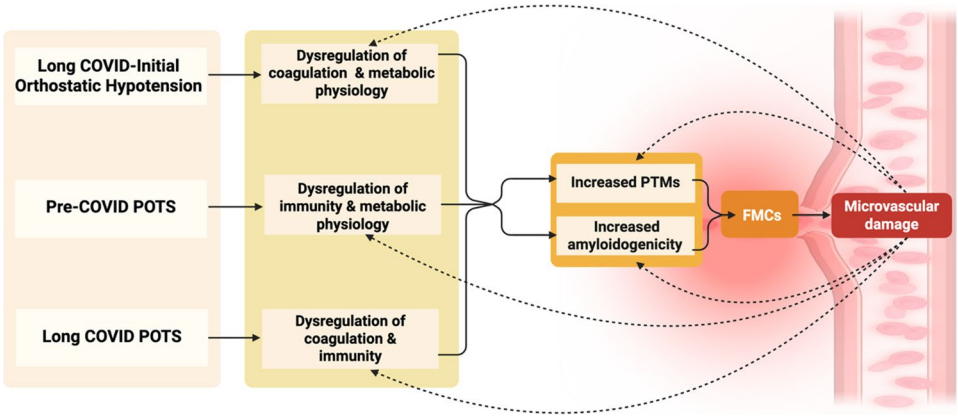



**Fig. 5** A schematic figure summarizing the proposed mechanistic relationship between PTMs, fibrinaloid microclot complexes (FMCs), coagulation, immunity, and metabolic physiology phenotypes

diabetic PTM patterns. Future work could include a data-dependant analysis supplementary to the data-independent analysis and a focus on the functional implications of specific PTM's in relation to the protein and the modification site of the protein.

## Conclusion

This study provides the first direct evidence that PTMs within FMC rich PPP-IF has the potential to distinctly characterize PC-POTS, LC-IOH, and LC-POTS. Since the PC-POTS samples predate the COVID-19 pandemic, the PTM patterns represent intrinsic POTS features rather than SARS-CoV-2 consequences, enabling the contributions of classical POTS biology and Long COVID dysregulation to be disentangled (Fig. 5).

PTM profiling reveals distinct yet overlapping biochemical signatures invisible to standard blood tests: LC-IOH exhibits pro-coagulant fibrinogen modifications with metabolic-like dysregulation. PC-POTS and LC-POTS shows predominantly immune-oxidative disruptions, with PC-POTS also showing metabolic dysregulation and LC-POTS include oxidation-modifications effecting coagulation. These PTM landscapes highlight potential avenues for biomarker-based diagnosis, patient stratification, and therapeutic strategies targeting glycation, oxidative stress, and complement pathways.

The consistent presence of amyloidogenic peptides across groups underscores FMCs within PPP-IF as β-sheet-rich, fibrinolysis-resistant aggregates contributing to microvascular dysfunction. Minimal protein-abundance differences alongside substantial PTM dysregulation suggest a possible mechanistic explanation for severe symptoms despite normal routine lab findings.

Overall, these findings offer a unifying framework and the first provable hypothesis for shared and divergent mechanisms in post-viral and dysautonomic conditions, establishing a novel and foundational avenue for biomarker-research, which could lead to biomarker-driven diagnostics and personalized therapeutics across the spectrum of autonomic and post-viral disorders.

**Abbreviations**

| | |
|---|---|
| A2MG | Alpha-2-macroglobulin |
| AGEs | Advanced glycation end-products |
| ALBU | Albumin |
| AmBic | Ammonium bicarbonate |
| ANT3 | Antithrombin-III |
| APOA1 | Apolipoprotein A-I |
| APOB | Apolipoprotein B |
| BMI | Body mass index |
| BSA | Body surface area |
| CFS | Chronic Fatigue Syndrome |
| CHREB | Conjoint Health Research Ethics Board |
| CO3 | Complement C3 |
| CO4/B | Complement C4-A and C4-B |
| DIA | Data-independent acquisition |
| FDR | False discovery rate |
| FIBA, FIBB, FIBG | Fibrinogen alpha, beta, gamma chains |
| FMCs | Fibrinaloid microclot complexes |
| HBA, HBB | Hemoglobin alpha and beta |
| HDL | High density lipoprotein |
| IC1 | C1 inhibitor |
| IGG1 | immunoglobulin gamma-1 |
| ITIH4 | Inter-alpha-trypsin inhibitor heavy chain H4 |
| LC-IOH | Long COVID with initial orthostatic hypotension |
| LC-MS/MS | Liquid chromatography-tandem mass spectrometry |
| LC-POTS | Long COVID POTS |
| MCAS | Mast Cell Activation Syndrome |
| MeOH | Methanol |
| MMTS | S-methyl methanethiosulfonate |
| MTA | Material Transfer Agreement |
| MYH9 | Myosin-9 |
| NETs | Neutrophil extracellular traps |
| PASC | Post Acute COVID-19 sequelae |
| PASEF | Parallel accumulation-serial fragmentation |
| PBS | Phosphate-buffered saline |
| PC-POTS | Pre-COVID-19 POTS |
| PEM | Post-exertional Malaise |
| POTS | Postural Orthostatic Tachycardia Syndrome |
| PPP | Platelet-poor plasma |
| PPP-IF | Insoluble fraction of platelet-poor plasma |
| PTMs | Post-translational modifications |
| PYD | Pyroglutamate |
| T2DM | Type 2 Diabetes Mellitus |
| TCEP | Tris(2-carboxyethyl) phosphine |
| TFA | Trifluoroacetic acid |
| ZA2G | Zinc-alpha-2-glycoprotein |

## Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12933-026-03296-9.

Supplementary Material 1.

**Acknowledgements**
This work was funded by StandingUpToPOTS 2024/2025: Pretorius, Kell and Lip grant.

**Authors' contributions**
Methodology: RB, EP, MV, SRR, RHSample acquisition : RH, SRRInvestigation : RB, EP, MVVisualisation and analyses: RB, CB, MVFunding acquisition: EP, DKProject administration: RB, MV, SRR, RHWriting original draft: RB, EPWriting, review and editing: RB, EP, CB, MV, SR, RH, DK, AK.

**Data availability**

**Lead Contact**
Requests for further information, resources, or data should be directed to and will be fulfilled by the lead contacts, Etheresia Pretorius (resiap@sun.ac.za) and Renata M. Booyens (renatab@sun.ac.za).

**Materials Availability**
This study did not generate new unique reagents or materials.

**Data and Code Availability**
The study generated mass spectrometry-based proteomics datasets, including raw and processed DIA-PASEF LC-MS/MS data. Additional information required to reanalyse the data reported in this study is available from the lead contacts upon reasonable request.

## Declarations

**Declaration of interests**
EP is an author of a patent: NEW METHOD TO DIAGNOSE INFLAMMATORY DISEASES 11194720 PCT application number PCT/EP2022/072147, Date of receipt 05 August 2022. EP is an author of a patent DIAGNOSTIC METHOD FOR LONG COVID PCT application number GB2105644.5. EP is a founding director of Biocode Technologies, a Stellenbosch University start-up company. RB is a postgraduate student of EP.
SRR has consultancies with Regeneron, argenx BV, Antag Pharma, Novo Nordisk, and Lumia Health; grants from the Heart & Stroke Foundation of Canada, Dysautonomia International, StandingUpToPOTS, and the Canerector Foundation; and research trial enrolment fees from Medtronic Inc. MAK has appeared on the StandingUpToPOTS podcast as a patient/clinician contributor; no honorarium was received. None of these relationships are related to the present work. All other authors: no competing interests to declare.

**Author details**
[1]Department of Physiological Sciences, Stellenbosch University, Private Bag Z1, Stellenbosch 7599, South Africa
[2]Trace Labs, Cape Town, South Africa
[3]Department of Genetics, Stellenbosch University, Private Bag Z1, Stellenbosch 7599, South Africa
[4]Libin Cardiovascular Institute, University of Calgary, Calgary, CA, Canada
[5]Pulmonologist and sleep physician (medically retired), Manchester, UK
[6]Department of Biochemistry, Cell and Systems Biology, Institute of Systems, Molecular and Integrative Biology, University of Liverpool, Liverpool L69 7ZB, UK

Received: 14 May 2026 / Accepted: 8 July 2026
Published online: 23 July 2026

**References**

1. Dharmindra Dulal D, Maraey A, Elsharnoby H, Chacko P, Grubb B. Impact of COVID-19 pandemic on the incidence and prevalence of postural orthostatic tachycardia syndrome. Eur Heart J Qual Care Clin Outcomes. 2025;11(5):698–704. https://doi.org/10.1093/ehjqcco/qcae111.
2. Yong SJ. Long COVID or post-COVID-19 syndrome: putative pathophysiology, risk factors, and treatments. Infect Dis. 2021;53(10):737–54. https://doi.org/10.1080/23744235.2021.1924397.
3. Mandel H et al. Jul., Long COVID Incidence Proportion in Adults and Children Between 2020 and 2024: An Electronic Health Record-Based Study From the RECOVER Initiative., Clin. Infect. Dis., vol. 80, no. 6, pp. 1247–1261, 2025, https://doi.org/10.1093/cid/ciaf046.
4. B. B et al., 2022 AHA/ACC key data elements and definitions for cardiovascular and noncardiovascular complications of COVID-19. J Am Coll Cardiol, 80, 4, pp. 388–465, 2022, https://doi.org/10.1016/j.jacc.2022.03.355.
5. Fedorowski A. Postural orthostatic tachycardia syndrome: clinical presentation, aetiology and management. J Intern Med. Apr. 2019;285(4):352–66. https://doi.org/10.1111/joim.12852.
6. Vernino S, et al. Postural orthostatic tachycardia syndrome (POTS): State of the science and clinical care from a 2019 National Institutes of Health Expert Consensus Meeting - Part 1. Auton Neurosci. 2021;235:102828. https://doi.org/10.1016/j.autneu.2021.102828.
7. Raj SR, et al. Postural orthostatic tachycardia syndrome (POTS): Priorities for POTS care and research from a 2019 National Institutes of Health Expert Consensus Meeting - Part 2. Auton Neurosci. 2021;235:102836. https://doi.org/10.1016/j.autneu.2021.102836.
8. Shaw BH, et al. The face of postural tachycardia syndrome - insights from a large cross-sectional online community-based survey. J Intern Med. 2019;286(4):438–48. https://doi.org/10.1111/joim.12895.
9. DePace NL, Colombo J. Long-COVID syndrome and the cardiovascular system: A review of neurocardiologic effects on multiple systems. Curr Cardiol Rep. 2022;24:1711–26. https://doi.org/10.1007/s11886-022-01786-2.
10. D B, P. M, B. AL, and, JA B, Long COVID. A proposed hypothesis-driven model of viral persistence for the pathophysiology of the syndrome, Allergy Asthma Proc., vol. 43, no. 3, pp. 187–193, 2022, https://doi.org/10.2500/aap.2022.43.220018.
11. Lesley K, Pract BJG. Br J Gen Pract. 2022;72(714):8–9. https://doi.org/10.3399/bjgp22X718037.
12. Johansson M, Yan H, Welinder C, others. Plasma proteomic profiling in postural orthostatic tachycardia syndrome (POTS) reveals new disease pathways. Sci Rep. 2022;12:20051. https://doi.org/10.1038/s41598-022-24729-x.
13. Pretorius E, Venter C, Laubscher GJ, Lourens PJ, Steenkamp J, Kell DB. Prevalence of readily detected amyloid blood clots in 'unclotted' Type 2 Diabetes Mellitus and COVID-19 plasma: a preliminary report. Cardiovasc Diabetol. 2020;19:193. https://doi.org/10.1186/s12933-020-01165-7.
14. Pretorius E, et al. Persistent clotting protein pathology in Long COVID/Post-Acute Sequelae of COVID-19 (PASC) is accompanied by increased levels of antiplasmin. Cardiovasc Diabetol. 2021;20:172. https://doi.org/10.1186/s12933-021-01359-7.
15. Pretorius E, Venter C, Nunes JM, Thierry AR, Kell DB. Inflammatory Triggers, Cell Death, Membrane Damage and Lipid Asymmetry That Shape Procoagulant Surfaces for Amyloidogenic Microclotting. Oct. 2025;21. https://doi.org/10.20944/preprints202510.1460.v1.
16. Thierry AR et al. Oct., Circulating Microclots Are Structurally Associated With Neutrophil Extracellular Traps and Their Amounts Are Elevated in Long COVID Patients, J. Med. Virol., vol. 97, no. 10, 2025, https://doi.org/10.1002/jmv.70613.
17. Kell DB, Laubscher GJ, Pretorius E. A central role for amyloid fibrin microclots in long COVID/PASC: origins and therapeutic implications, Biochemical Journal, vol. 479, no. 4, pp. 537–559, Feb. 2022, https://doi.org/10.1042/BCJ20220016.
18. B. G et al., Post-Acute Sequelae of COVID-19 and Cardiovascular Autonomic Dysfunction: What Do We Know? J Cardiovasc Dev Dis, 8, 11, p. 156, 2021, https://doi.org/10.3390/jcdd8110156.
19. Johansson M, et al. Long-Haul Post-COVID-19 Symptoms Presenting as a Variant of Postural Orthostatic Tachycardia Syndrome: The Swedish Experience. JACC Case Rep. 2021;3(4):573–80. https://doi.org/10.1016/j.jaccas.2021.01.009.
20. Kanjwal K, Jamal S, Kichloo A, Grubb BP. New-onset Postural Orthostatic Tachycardia Syndrome Following Coronavirus Disease 2019 Infection. J Innov Card Rhythm Manag. 2020;11(11):4302–4. https://doi.org/10.19102/icrm.2020.111102.
21. Gunning WT 3, Khan S, Spatafore JW, Karabin BL, Grubb BP. Postural orthostatic tachycardia syndrome in post-COVID-19 long-hauler patients is associated with platelet storage pool deficiency. Front Med (Lausanne). 2025;12:1560120. https://doi.org/10.3389/fmed.2025.1560120.
22. Kell DB, Khan MA, Kane B, Lip GYH, Pretorius E. Possible role of fibrinaloid microclots in postural orthostatic tachycardia syndrome (POTS): Focus on long COVID. J Pers Med. 2024;14(2). https://doi.org/10.3390/jpm14020170.

23. Kruger A, Vlok M, Turner S, others. Proteomics of fibrin amyloid microclots in long COVID/post-acute sequelae of COVID-19 (PASC) shows many entrapped pro-inflammatory molecules that may also contribute to a failed fibrinolytic system. Cardiovasc Diabetol. 2022;21:190. https://doi.org/10.1186/s12933-022-01623-4.
24. Thomas C, Faghy MA, Chidley C, Phillips BE, Bewick T, Ashton RE. Blood Biomarkers of Long COVID: A Systematic Review, Mol. Diagn. Ther., vol. 28, no. 5, pp. 537–574, Sep. 2024, https://doi.org/10.1007/s40291-024-00731-z.
25. Dutta H, Jain N. Post-translational modifications and their implications in cancer. Front Oncol. 2023;13:1240115. https://doi.org/10.3389/fonc.2023.1240115.
26. Nencini F, et al. Post-translational modifications of fibrinogen: implications for clotting, fibrin structure and degradation. Mol Biomed. 2024;5(1):45. https://doi.org/10.1186/s43556-024-00214-x.
27. Mikuteit M et al. Sep., Amino acids, post-translational modifications, nitric oxide, and oxidative stress in serum and urine of long COVID and ex COVID human subjects., Amino Acids, vol. 55, no. 9, pp. 1173–1188, 2023, https://doi.org/10.1007/s00726-023-03305-1.
28. Kell DB, Pretorius E. The Proteome Content of Blood Clots Observed Under Different Conditions: Successful Role in Predicting Clot Amyloid(ogenicity), Molecules, vol. 30, no. 3, p. 668, Feb. 2025, https://doi.org/10.3390/molecules30030668.
29. Kell DB, Pretorius E. Proteomic Evidence for Amyloidogenic Cross-Seeding in Fibrinaloid Microclots. Int J Mol Sci. Oct. 2024;25(19):10809. https://doi.org/10.3390/ijms251910809.
30. Kell DB, Doyle KM, Salcedo-Sora JE, Sekhar A, Walker M, Pretorius E. AmyloGram reveals amyloidogenic potential in stroke thrombus proteomes, Biochemical Journal, vol. 482, no. 22, pp. 1689–1706, Nov. 2025, https://doi.org/10.1042/BCJ20253317
31. Grixti JM, et al. Amyloid Presence in Acute Ischemic Stroke Thrombi: Observational Evidence for Fibrinolytic Resistance. Stroke. Jul. 2025;56(7). https://doi.org/10.1161/STROKEAHA.124.050033.
32. Freeman R, et al. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. Clin Auton Res. Apr. 2011;21(2):69–72. https://doi.org/10.1007/s10286-011-0119-5.
33. Raj SR, et al. Canadian Cardiovascular Society Position Statement on Postural Orthostatic Tachycardia Syndrome (POTS) and Related Disorders of Chronic Orthostatic Intolerance. Can J Cardiol. 2020;36(3):357–72. https://doi.org/10.1016/j.cjca.2019.12.024.
34. Zhou Y, et al. Metascape provides a biologist-oriented resource for the analysis of systems-level datasets. Nat Commun. Apr. 2019;10(1):1523. https://doi.org/10.1038/s41467-019-09234-6.
35. Nencini F, Giurranna E, Borghi S, Taddei N, Fiorillo C, Becatti M. Fibrinogen oxidation and thrombosis: shaping structure and function. Antioxid (Basel). 2025;14(4):390. https://doi.org/10.3390/antiox14040390.
36. Twarda-Clapa A, Olczak A, Białkowska AM, Koziołkiewicz M. Advanced glycation end-products (AGEs): Formation, chemistry, classification, receptors, and diseases related to AGEs. Cells. 2022;11(8):1312. https://doi.org/10.3390/cells11081312.
37. B. AV et al., Oxidation-induced modification of the fibrinogen polypeptide chains. Dokl Biochem Biophys, 474, 1, pp. 173–7, 2017, https://doi.org/10.1134/S1607672917030115
38. Pretorius E, Venter C, Nunes JM, Thierry AR, Kell DB. Inflammatory Triggers, Cell Death, Membrane Damage and Lipid Asymmetry. That Shape Procoagulant Surfaces for Amyloidogenic Microclotting; 2025.
39. Stief T, Aab A, Heimburger N. Oxidative inactivation of purified human alpha-2-antiplasmin, antithrombin III, and C1-inhibitor. Thromb Res. 1988;49:581–9. https://doi.org/10.1016/0049-3848(88)90255-1.
40. Yan B, Hu D, Knowles S, Smith J. Probing chemical and conformational differences in the resting and active conformers of platelet integrin alpha(IIb) beta(3). J Biol Chem. 2000;275(10):7249–60.
41. West EE, Kolev M, Kemper C. Complement and the Regulation of T Cell Responses. Annu Rev Immunol. Apr. 2018;36:309–38. https://doi.org/10.1146/annurev-immunol-042617-053245.
42. Ali Komi DE, Shafaghat F, Kovanen PT, Meri S. Mast cells and complement system: Ancient interactions between components of innate immunity, Allergy, vol. 75, no. 11, pp. 2818–2828, Nov. 2020, https://doi.org/10.1111/all.14413
43. Kohno R, et al. Mast Cell Activation Disorder and Postural Orthostatic Tachycardia Syndrome: A Clinical Association. J Am Heart Assoc. Sep. 2021;10(17). https://doi.org/10.1161/JAHA.121.021002.
44. Verhelst J, Impens Y. Alpha-2-macroglobulin in inflammation, immunity and infections. Front Immunol. 2021;12:803244. https://doi.org/10.3389/fimmu.2021.803244.
45. Vandooren J, Itoh Y. Alpha-2-macroglobulin in inflammation, immunity and infections. Front Immunol. 2021;12:803244. https://doi.org/10.3389/fimmu.2021.803244.
46. Jing D, Jin J, Mei Z, Zhu Q, Lu Y, Wang X. Effects of Helicobacter pylori infection and interleukin 6 on the expression of ITIH4 in human gastric cancer cells. Transl Cancer Res. 2020;9:4656–65. https://doi.org/10.21037/tcr-20-1766.
47. CJ B. Apolipoprotein B-containing lipoproteins in atherogenesis. Nat Rev Cardiol. 2025;22:399–413. https://doi.org/10.1038/s41569-024-01111-0.
48. Chevalier C, et al. Post-translational modifications of apolipoproteins as promising biomarkers for diabetes-related cardiovascular diseases: A comprehensive review. Diabetes Metab. Sep. 2025;51(5):101683. https://doi.org/10.1016/j.diabet.2025.101683.
49. Yang M, et al. Zinc-α2-glycoprotein is associated with insulin resistance in humans and is regulated by hyperglycemia, hyperinsulinemia, or liraglutide administration: cross-sectional and interventional studies in normal subjects, insulin-resistant subjects, and subjects with newly diagnosed diabetes. Diabetes Care. 2013;36(5):1074–82. https://doi.org/10.2337/dc12-0940.
50. È, Navarro-Masip, et al. Acute downregulation of zinc α2-glycoprotein: Evidence from human and HepG2 cell studies. Int J Mol Sci. 2025;26(12):5438. https://doi.org/10.3390/ijms26125438.
51. Bramante CT, et al. Outpatient treatment of COVID-19 and incidence of post-COVID-19 condition over 10 months (COVID-OUT): a multicentre, randomised, quadruple-blind, parallel-group, phase 3 trial. Lancet Infect Dis. Oct. 2023;23(10):1119–29. https://doi.org/10.1016/S1473-3099(23)00299-2.


## Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.